A Digital Twin for Shading-Aware Assessment of Vehicle-Integrated Photovoltaics on a Heavy Electric Vehicle

Aditya Kumar, Anandsagar Gaikwad, and Arjun Deo

***Abstract*—Feasibility studies of vehicle-integrated photovoltaics (VIPV) usually treat the whole solar skin as a horizontal roof that is never shaded. We extend SOLARIS, an hour-by-hour energy-balance toolkit for a $2.5\,\mathrm{t}$ seven-seat electric vehicle, with a digital twin of central Phoenix, Arizona. The twin builds a $0.5\,\mathrm{m}$ city model from 5.28 billion lidar points, drives the vehicle on traffic-simulated trips along three arterials, parks it at five sites and computes the shadow on each flat facet of an assumed $11.6\,\mathrm{m}^2$ skin every second while driving and every five minutes while parked. On a satellite-derived typical year the conference model gives a solar self-sufficiency of 0.832 at $50\,\mathrm{km/day}$. Facet orientation lowers it to 0.677. After parking shade, driving shade and electrical mismatch, a suburban commuter keeps 0.512 and a car parked at a downtown curb 0.256; most of the shading loss occurs while parked. A shading-aware design equation stays within 2.6% of the full simulation, and a Unity twin agrees with the Python shading engine on 99.1% of 15.5 million sample-seconds. The code is open source.**

***Index Terms*—Digital twin, electric vehicles, energy self-sufficiency, urban shading, vehicle-integrated photovoltaics.**

NOMENCLATURE

f_{ss}	Self-sufficiency, harvest over demand.
A, A_{req}	PV area; area for $f_{\mathrm{ss}} = 1$ (m^2).
D	Daily distance (km).
c	Battery-side consumption (Wh/km).
E_{aux}	Daily auxiliary energy (kWh).
η_{pv}, κ_{pv}	Cell efficiency; system derating.
PSH, f_T	Peak-sun hours; temperature factor.
G_f	Irradiance on facet f ($\mathrm{W/m}^2$).
$\bar{v}_{b,f}$, $\bar{v}_{h,f}$	Beam visibility; horizon-band view.
SVF_f	Cosine-weighted sky view of facet f.
τ, s	Tree transmissivity; shaded fraction.
$F_{\mathrm{geo}}, F_{\mathrm{sh}}, F_{\mathrm{mm}}$	Orientation, shading, mismatch factors.
F_{park}	Parked harvest relative to open sky.

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This work extends the paper "SOLARIS: A first-principles, drive-cycle- and climate-resolved methodology and open toolkit for assessing solar-assisted heavy electric vehicles," accepted for presentation at the IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC-AP 2026), Busan, Republic of Korea, 1–4 December 2026 [1].

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I. INTRODUCTION

SOLAR cells built into the body of an electric vehicle (EV), known as vehicle-integrated photovoltaics (VIPV), can supply part of the energy it would otherwise draw from the grid. How large that part is depends on sunlight, consumption and their timing. In an earlier paper [1] we built SOLARIS, an open toolkit [2] that charges every harvesting component to the vehicle's mass, computes consumption on the Worldwide harmonized Light vehicles Test Procedure (WLTP) cycle and steps the battery through a year hour by hour. For a $2.5\,\mathrm{t}$ seven-seat vehicle with an $11.6\,\mathrm{m}^2$ skin, satellite-derived weather gave a solar self-sufficiency f_{ss} of 0.83 in Phoenix, 0.71 in Lucknow and 0.52 in London for $50\,\mathrm{km/day}$ of urban driving.

Those numbers rest on two simplifications that most VIPV feasibility studies share. The whole skin receives global horizontal irradiance (GHI), although the sides of the vehicle are nearly vertical. Nothing shades the vehicle, although a car is parked for most of the day, often next to buildings or under trees, and drives through street canyons. Both make the result optimistic, and the size of the error depends on where the car is driven and parked. The conference paper named route-level urban shading as its planned extension [1].

This paper carries out that extension. A digital twin of central Phoenix replaces both simplifications and keeps the rest of the conference model unchanged, so each correction can be measured on its own. The contributions are:

1) a reproducible pipeline that turns public lidar, building and street data into a $0.5\,\mathrm{m}$ city model, with buildings and trees kept separate, the ground checked against survey marks and later building changes corrected;
2) a facet-by-facet model (a facet is one flat PV panel of the skin) of sunlight, cell temperature and electrical output, run every second while the car drives in simulated traffic and every five minutes while it is parked, with the energy balance checked at every step;
3) for four daily uses of the car, a year-long breakdown of the self-sufficiency lost to facet orientation, parking shade, driving shade and electrical mismatch;
4) a shading-aware version of the conference design equation and a design chart, checked against the full simulation;
5) an independent check of the shading engines against a Unity ray-casting twin, the SOLWEIG urban-climate model and street-level photographs.

II. Related Work

Reviews of VIPV agree that yield is set by geometry, curvature, shading and climate [3]–[5]. Measurements with a pyranometer array on a moving car showed how unevenly the roof and sides are lit [6], and three-dimensional irradiance models for curved car bodies followed [7]. Yield models compute harvest for fleets [8] or along shaded city routes [9], but take consumption as given. From geographic data, Brito *et al.* [10] found annual shading losses of up to 25% on Lisbon roads and over 50% at urban parking spaces, and Hoth *et al.* [11] report a median shading loss of 61% (quartiles 38% and 76%) for parked cars in Berlin.

Building-scale solar potential is routinely computed from lidar surface models with hourly sky and shadow calculations [12]. In urban climate work, the SOLWEIG model in the Urban Multi-scale Environmental Predictor (UMEP) [13], [14] is a standard reference for shadow patterns, and sky view factors (SVF) for street routing in Phoenix have been computed from synthetic fisheye images [15]. These tools describe fixed surfaces; a vehicle moves through them and changes heading every second.

Partial shade reduces output more than the shaded area suggests, because shaded cells force bypass diodes to conduct [16]; distributed converters recover part of that loss [17]. On the vehicle side, studies optimize the PV converter and charger [18], [19] or quantify how cabin climate control changes consumption [20]. San José *et al.* [9] recorded shadows on a driving car with a camera and showed how partial shading degrades the current–voltage curve. To our knowledge no study has combined trajectory-level driving and parking shade, facet orientation and electrical mismatch with drive-cycle demand and an annual battery simulation, or done so for a heavy vehicle.

III. Conference Baseline

The conference model [1], [2] is used unchanged as the reference (version 1.1.0; the refactored code reproduces every output of the conference code, as the repository tests check). Demand is the WLTP consumption at the battery, $210\,\mathrm{Wh/km}$ on the city part of the cycle and $237\,\mathrm{Wh/km}$ on the full cycle, plus auxiliaries and cabin cooling. Harvest is

$$E_{\mathrm{pv}} = A\,\eta_{\mathrm{pv}}\,\kappa_{\mathrm{pv}}\,\frac{G}{1000}\,[1+\beta(T_c-25)] \tag{1}$$

with $A = 11.6\,\mathrm{m}^2$, $\eta_{\mathrm{pv}} = 0.22$, $\kappa_{\mathrm{pv}} = 0.80$, $\beta = -0.4\%/\mathrm{K}$, $T_c = T_a + 25G/1000$ and G the GHI, plus small thermoelectric and suspension terms. The state of charge (SoC) of the $250\,\mathrm{kWh}$ pack is simulated hour by hour over a periodic year and recharged from 10% to 80% when needed. Scenario II is $50\,\mathrm{km/day}$ on the city cycle and Scenario I $100\,\mathrm{km/day}$ on the full cycle, both with trips at 08:00 and 18:00, and f_{ss} is annual harvest over annual demand. Setting harvest equal to demand gives the design area

$$A_{\mathrm{req}} = \frac{D\,c + E_{\mathrm{aux}}}{\eta_{\mathrm{pv}}\,\mathrm{PSH}\,\kappa_{\mathrm{pv}}\,f_T}, \tag{2}$$

where D is the daily distance, c the battery-side consumption, PSH the daily peak-sun hours, f_T the irradiance-weighted temperature factor and E_{aux} the daily auxiliary energy.

TABLE I
Conference Baseline on Satellite-Derived Hourly TMY

City	Scen.	f_{ss}	Charges/yr (no solar)	Relief (%)	A_{req} (m^2)	$D_{\max}$ (km/day)
Phoenix	II	0.832	5 (27)	82	14.4	37.5
	I	0.419	31 (54)	42	28.8	33.6
Lucknow	II	0.713	8 (27)	71	16.9	29.6
	I	0.363	35 (53)	36	33.4	26.6
London	II	0.517	12 (24)	52	23.9	20.5
	I	0.248	39 (50)	25	50.2	18.7

Relief: harvest used (harvest minus curtailment) over demand. $D_{\max}$: daily distance that $11.6\,\mathrm{m}^2$ makes self-sufficient. TMY annual GHI 2110.0, 1843.6 and 1050.7 $\mathrm{kWh/m}^2$.

Table I is the baseline (B0) on satellite-derived hourly typical meteorological years (TMY): National Solar Radiation Database (NSRDB) data for Phoenix and Lucknow [21] and Photovoltaic Geographical Information System (PVGIS) data for London [22]. Everything that follows is for Phoenix, Scenario II unless stated, starting from $f_{\mathrm{ss}} = 0.832$, 5 grid charges per year (27 without solar) and $A_{\mathrm{req}} = 14.4\,\mathrm{m}^2$. The idealized synthetic climate of the conference paper gives 0.94; of the gap of 0.11, 0.06 comes from the annual irradiation and 0.05 from the timing of cloud.

IV. Digital Twin

Fig. 1 shows the twin. Python computes all physics; Unity replays the same trips against the same geometry and is used only to check visibility (Section VIII). A Snakemake workflow [23] builds every number and figure in this paper from the raw data, with fixed seeds.

A. City Model

The area of interest is a $238\,\mathrm{km}^2$ box around the routes and parking sites (Fig. 2). U.S. Geological Survey (USGS) 3D Elevation Program (3DEP) lidar flown between October 2020 and December 2021 [24] gives 5.28 billion points after noise removal, 22.2 points/m^2. Each lidar return is assigned to the $0.5\,\mathrm{m}$ grid cell it falls in, giving a digital surface model (DSM), a ground model (DEM) and a canopy height model; 23.9% of the area is building and 4.4% tree. Buildings and trees are separated with 2021 National Agriculture Imagery Program (NAIP) imagery [25] and Overture building footprints [26]. Two corrections were needed. In aerial images tall buildings lean outward, so tower walls can look like vegetation in NAIP; pixels taller than $35\,\mathrm{m}$ are therefore never classed as trees. Power lines, poles and signal arms formed thin walls along the arterials and shaded even the most open route at noon, so height-only structures narrower than $1.5\,\mathrm{m}$ are removed. Against 87 survey marks of the National Geodetic Survey the DEM has a median offset of −0.11 m (median absolute deviation 0.14 m). Of 159 buildings that differ between the lidar and current footprints along the routes, the largest 45 were checked against current imagery: 14 added (among them a 23-floor tower on the dense route), 12 removed and 19 rejected as bridges, decks or trees. OpenStreetMap (OSM) [27] bridges and tunnels mark covered road. Simplified triangle

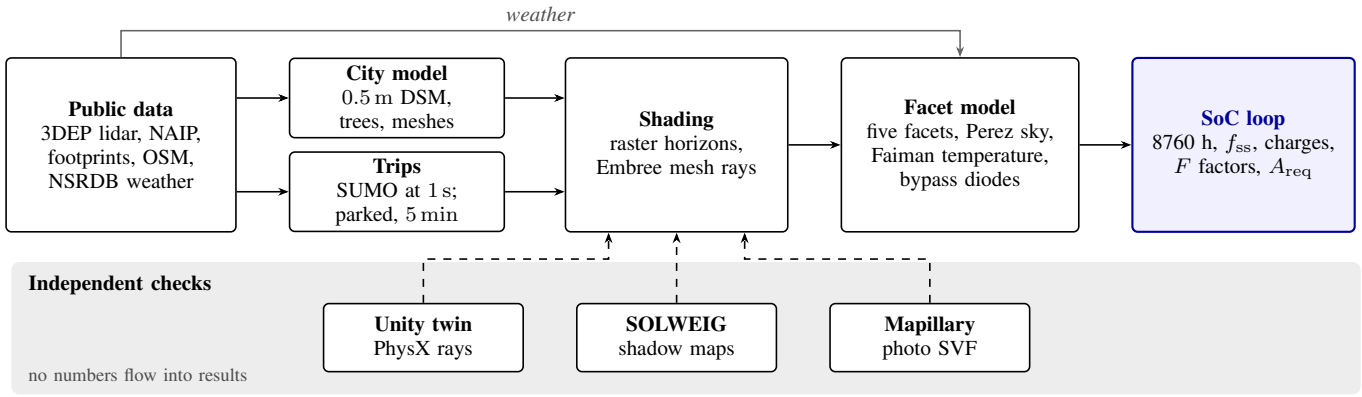


Fig. 1. Model chain of the twin (top) and the independent checks (bottom). Python computes every energy value; the checks compare only visibility or sky view with the shading engines.

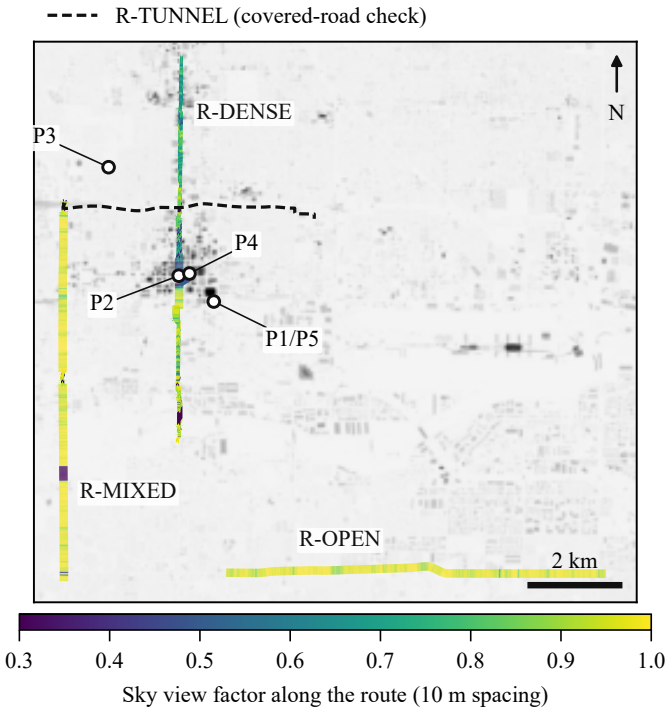


Fig. 2. Study area: building and tree height above ground (gray), the three routes colored by the sky view factor used to select them, the covered-road check route and the parking sites (UTM zone 12N). P5 is the P1 stall under a modeled shade roof.

meshes of the DSM and of the trees feed the mesh engine and Unity.

B. Routes, Traffic and Parking Sites

Routes were chosen by a fixed rule from 128 candidate windows, 8 km long along the axis of the named arterials (the driven lines are 7.8 to 8.3 km): the highest mean SVF (R-OPEN, Southern Avenue, SVF 0.96), the lowest (R-DENSE, Central Avenue through downtown, 0.74) and the highest SVF variance in the middle band (R-MIXED, 19th Avenue, 0.90). A fourth route, R-TUNNEL through the I-10 Deck Park Tunnel, tests covered road only. Simulation of Urban Mobility (SUMO) [28] on an OpenStreetMap network produced 120 ego traces at 1 Hz, 30 per route: both directions, five departures from 08:00 to 16:00 and three traffic demands. Each trace is driven on four dates, 21 March, 21 June, 21 September and 21 December. The five parking sites (Table II) are an open lot (P1), a curb in the downtown canyon (P2), a residential driveway beside a tree (P3), a garage roof deck (P4) and the P1 stall under a $4\,\mathrm{m}\times3\,\mathrm{m}$ shade roof (P5).

C. Shading Engines

The raster engine computes, for every sample point, the horizon in 360 azimuths out to 1 km, once over the building surface (h^{b}) and once over buildings and trees (h^{bt}), with azimuths corrected for meridian convergence. A ray from each facet sample toward the sun decides its beam state at every time step, and the mean over the N_f samples of facet f is

$$\bar{v}_{b,f} = \frac{1}{N_f}\sum_{i\in f}\begin{cases}1 & \text{ray clears buildings and trees,}\\ \tau & \text{ray blocked by trees only,}\\ 0 & \text{ray blocked by a building,}\end{cases} \tag{3}$$

with $\tau = 0.03$. The sky terms use the horizon at the roof center. With $w_f(\omega) = \max(0, \mathbf{n}_f \cdot \omega)$ the cosine weight of sky direction ω for a facet with normal $\mathbf{n}_f$, and $S_f(h) = \int_{\omega>h} w_f\, d\Omega \big/ \int_{\Omega} w_f\, d\Omega$ the weighted share of the sky above horizon h,

$$\mathrm{SVF}_f = S_f(h^{\mathrm{bt}}) + \tau\left[S_f(h^{\mathrm{b}}) - S_f(h^{\mathrm{bt}})\right], \tag{4}$$

integrated on a $1^\circ\times0.5^\circ$ sky grid; the horizon-band view $\bar{v}_{h,f}$ is the same share over the lowest 6° of sky. The mesh engine casts Embree rays [29] against the meshes. Sun positions come from the solar position algorithm (SPA) of the National Renewable Energy Laboratory (NREL) [30], as implemented in pvlib [31]. The two engines and SOLWEIG were checked in 40 tests covering sun position, transposition, shadow length and canyon SVF (Section VIII).

V. Vehicle Skin and Energy Chain

The intended skin layout of the vehicle is not fixed, so we use an assumed design layout of five facets (Fig. 3) that totals the conference $11.6\,\mathrm{m}^2$: roof $4.4\,\mathrm{m}^2$ at 2°, hood $1.6\,\mathrm{m}^2$ at 10°, tailgate $0.6\,\mathrm{m}^2$ at 60° and two sides of $2.5\,\mathrm{m}^2$ at 85°.

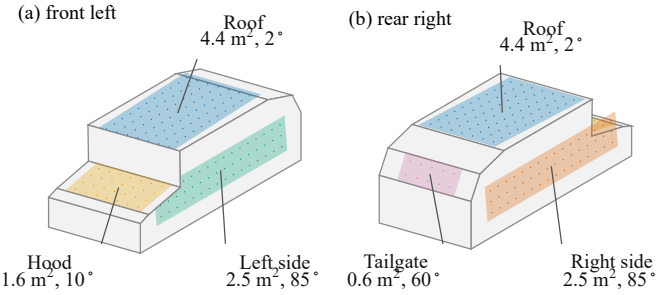


Fig. 3. Assumed design layout of the $11.6\,\mathrm{m}^2$ skin on a plain body, with the $0.25\,\mathrm{m}$ shading samples (area and tilt per facet). The hood shows in both views.

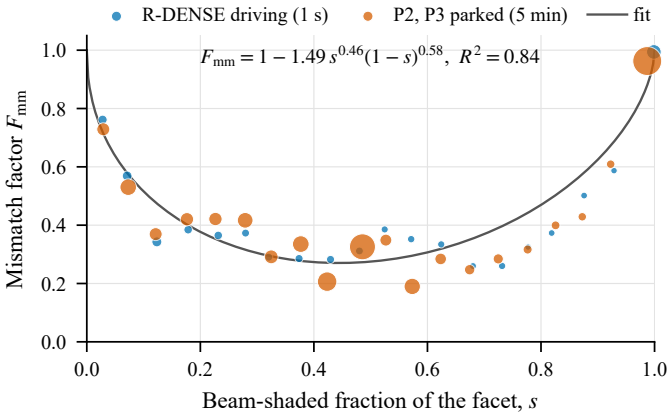


Fig. 4. Electrical mismatch factor against the beam-shaded fraction of a facet. Points are bin means; marker area is proportional to the energy in the bin. The line is the fit (6).

Like the mass audit in [1], this layout is a design estimate; results for the roof alone are given as a bound. Each facet is sampled on a $0.25\,\mathrm{m}$ grid. With the Perez model [32] split into beam (with circumsolar) G_b, isotropic G_d and horizon G_h components and ground reflection G_g (albedo 0.2), facet f receives

$$G_f = \bar{v}_{b,f}\,G_{b,f} + \mathrm{SVF}_f\,G_{d,f} + \bar{v}_{h,f}\,G_{h,f} + G_{g,f}; \qquad (5)$$

all three view terms are 1 in open sky, and all four terms are 0 on covered road. Cell temperature follows Faiman [33], $T_c = T_a + G_f/(u_0 + u_1 w)$ with pvlib's open-rack coefficients and w the wind plus the vehicle speed, and energy follows (1) with G_f for G.

Partial shade costs more electrically than its area suggests. Each facet is one module on its own maximum power point tracking (MPPT) input, its samples are cells, and the cells form three bypass-diode substrings. The current–voltage curves come from pvlib's implementation of Bishop's model [16] with reverse breakdown, using area-scaled parameters of a 22.1%-efficient back-contact cell. Under uniform light the string reproduces the averaged model within 7×10^{-5}. Over all R-DENSE trips and the P2 and P3 parking steps, the mismatch factor $F_{\mathrm{mm}} = P_{\mathrm{el}}/P_{\mathrm{avg}}$ is fitted against the beam-shaded fraction s of the facet (Fig. 4):

$$F_{\mathrm{mm}} = 1 - 1.49\,s^{0.46}(1-s)^{0.58}, \qquad R^2 = 0.84. \qquad (6)$$

One shaded cell already pushes its substring through the bypass diode, so F_{mm} drops steeply at small s.

Four daily-use archetypes keep the demand of their baseline scenario and change only where the car is when the sun shines. A1, a suburban commuter, drives R-MIXED and R-OPEN at 08:00 and 17:00 and parks at the open lot P1 by day and the driveway P3 at night. A2 and A3 drive R-DENSE and R-OPEN at the same hours and park downtown by day, at the canyon curb P2 (A2) or under the shade roof P5 (A3), and at P3 at night. A4, a fleet vehicle, drives $100\,\mathrm{km/day}$ on all three routes from 08:00 to 16:00, alternates P2 and P1 between trips and stays at P1 otherwise. On weekends the daytime parking moves to the night site. The evening trip runs at 17:00 instead of the conference's 18:00 because the traffic traces end with the 16:00 departure; it replays the 16:00 traces one hour later, and the "trip hours" step of the waterfall isolates this change. At each parking site, harvest is averaged over the directions in which a car can stand there. The $1\,\mathrm{s}$ driving steps and $5\,\mathrm{min}$ parking steps are summed into the 8760 hourly steps of the conference model. Hourly sums match the fine-step totals to a relative 1×10^{-14}, and every hour holds exactly $3600\,\mathrm{s}$ of driving or parking.

VI. Results

A. Facet Orientation

With every facet set horizontal, the new model reproduces the conference harvest to within −0.03% (target ±0.5%); Faiman cooling adds 0.55%. Averaged over headings, the roof and hood see 1.00 and 0.99 of GHI, the tailgate 0.74 and the sides 0.56. On the conference trip hours, facet orientation alone lowers f_{ss} from 0.832 to 0.678 and raises grid charges from 5 to 9 per year. The PV harvest falls to $F_{\mathrm{geo}} = 0.805$ of the horizontal skin with Faiman cooling ($f_{\mathrm{ss}} = 0.836$); the thermoelectric and suspension terms, 0.029 of demand, do not change. The roof alone would give 0.335.

B. Driving and Parking Shade

Table II lists the shading factors of the 360 trips and five sites on the TMY. While driving, buildings leave 0.93 of the open-sky harvest on R-OPEN and 0.72 on R-DENSE, and street trees take slightly more. Shading is strongest at the 08:00 and 16:00 departures, when the sun is low, and on R-DENSE the December noon trips also lose far more than the noon trips of the other dates (Fig. 5). On R-TUNNEL the covered segment receives no light, and the whole trip, covered for 62 of 199 s in one direction and 71 of 381 s in the other, keeps 0.67 and 0.79 of the open-sky energy. Parking spans a much wider range: annual F_{park} is 0.94 at the open lot, 0.45 beside the tree and 0.32 at the canyon curb, and the shade roof leaves 0.64. Against SOLWEIG the parked shadow maps agree on at least 96.7% of pixels. The driving losses on R-OPEN and R-MIXED stay within the 25% that Brito *et al.* [10] found for Lisbon roads, while R-DENSE, keeping only 0.71 of the open-sky energy, loses more; the curb and driveway losses lie between the Berlin quartiles of Hoth *et al.* [11].

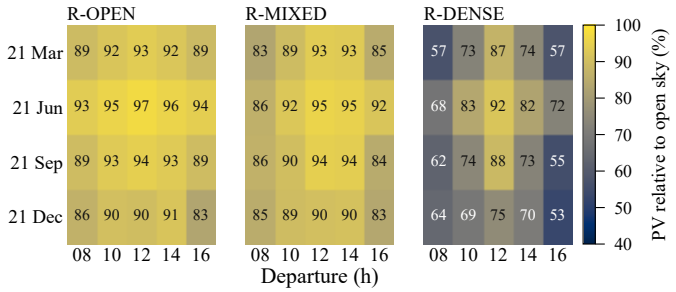


Fig. 5. Driving PV energy relative to open sky (%) by date and departure hour (buildings and trees, TMY, mean of both directions and three traffic demands).

TABLE II
ROUTES AND PARKING SITES: PV ENERGY RELATIVE TO OPEN SKY

Route	Length (km)	SVF	Buildings B/A	+ trees C/A	Beam shaded (%)
R-OPEN	8.3	0.96	0.93	0.91	10
R-MIXED	7.8	0.90	0.90	0.89	12
R-DENSE	7.8	0.74	0.72	0.71	32

Site	Setting	SVF	F_{park}
P1	open surface lot	0.95	0.94
P2	curb in the downtown canyon	0.34	0.32
P3	driveway beside a tree	0.48	0.45
P4	garage roof deck	0.91	0.88
P5	P1 stall under shade roof	0.95*	0.64

Driving: 360 trips on the TMY days (four dates, both directions, five departures, three demands), trees at $\tau = 0.03$; A open sky, B buildings, C buildings and trees. Parking: annual, averaged over headings. *SVF of the stall without the modeled roof.

C. Annual Loss Waterfall

Fig. 6 and Table III carry the baseline to each archetype one correction at a time. The archetype trip hours and the Perez–Faiman chain raise f_{ss} slightly (A1: 0.832 to 0.835). Facets then remove 0.159 for the 50 km archetypes, leaving 0.677; this differs from the 0.678 above only by the 17:00 return trip. Parking shade is the largest and most variable shading step: 0.125 for the commuter, 0.256 under the shade roof and 0.369 at the canyon curb, which is 92% to 95% of the shading loss. Driving past buildings costs 0.009 to 0.017, and street trees 0.001 or less. Electrical mismatch costs 0.030 to 0.067 for A1–A3 and 0.004 for A4. Energy-weighted, F_{mm} is 0.96 both on the R-DENSE trips and at the curb, where whole facets fall into building shadow, but 0.56 to 0.73 in the dappled shade of the tree at P3. The nights and weekends at P3 therefore pull the factors of A1–A3 in Table IV below the curb value, while A4, which never parks there, reaches 0.98. A3 loses more to mismatch than A2 (0.067 against 0.034), which differs only in parking at the curb instead of under the roof. In total the archetypes lose 38% (A1) to 69% (A2) of the baseline f_{ss}, and grid charges rise from 5 to between 13 and 20 a year for A1–A3 and from 31 to 41 for A4. Using 2023 weather instead of the TMY changes final f_{ss} by at most 0.004, and 5-min instead of hourly 2023 weather by at most 0.001, so hourly weather suffices for annual totals.

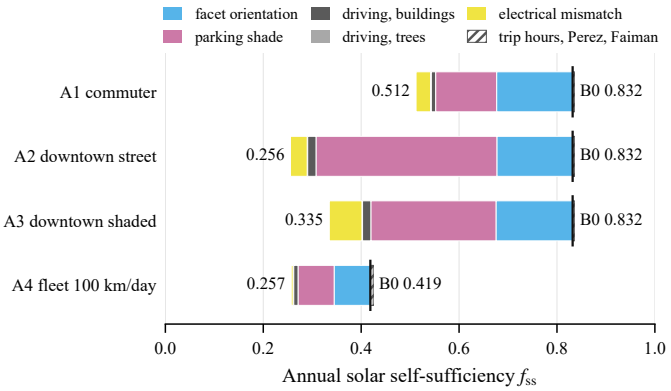


Fig. 6. Annual loss breakdown per archetype (Phoenix, TMY). Each bar runs from the final f_{ss} (left) to the baseline B0 (tick); colored segments are the losses by cause and the hatched segment is the small gain from trip hours and the Perez–Faiman chain.

TABLE III
ANNUAL RESULTS PER ARCHETYPE (PHOENIX, TMY)

	f_{ss}		Loss	Charges/yr	Relief (%)	Grid
Archetype	B0	final	(%)	B0→final	B0→final	(kWh/yr)
A1 commuter	0.832	0.512	38	5→13	82→51	2,300
A2 street	0.832	0.256	69	5→20	82→26	3,540
A3 shade roof	0.832	0.335	60	5→18	82→33	3,172
A4 fleet	0.419	0.257	39	31→41	42→26	7,251

A1–A3: 50 km/day, city cycle; A4: 100 km/day, full cycle. Loss: share of the B0 f_{ss}. Grid: final state.

VII. SHADING-AWARE DESIGN INVERSION

The waterfall suggests three factors that the conference design equation lacks. Each factor is the ratio of an archetype's annual harvest at two consecutive steps of the waterfall:

$$F_{\text{geo}} = \frac{E_{\text{facets}}}{E_{\text{flat}}}, \quad F_{\text{sh}} = \frac{E_{\text{shaded}}}{E_{\text{facets}}}, \quad F_{\text{mm}} = \frac{E_{\text{final}}}{E_{\text{shaded}}}, \tag{7}$$

where E_{flat} is the horizontal skin through the Perez–Faiman chain and E_{shaded} includes parking and driving shade, weighted by where the car actually is. Then

$$A_{\text{req}} = \frac{D\,c + E_{\text{aux}}}{\eta_{\text{pv}}\,\text{PSH}\,\kappa_{\text{pv}}\,f_T\,F_{\text{geo}}\,F_{\text{sh}}\,F_{\text{mm}}}. \tag{8}$$

To test it, the per-facet harvest of each archetype was recomputed and scaled to other areas and allocations (roof only, roof and hood, full skin, with the area spread over the allocated facets in proportion), and the hour-by-hour simulation, with the vehicle mass fed back at 3.5 kg/m^2, was solved for the area at which $f_{\text{ss}} = 1$. Over 60 cases (four archetypes, three allocations, $\kappa_{\text{pv}} = 0.60$ to 0.80) (8) stays within 2.6% of the simulation, inside the 3% target (3.3% if the mass is held fixed). Table IV lists the factors for the full skin and Fig. 7 the resulting design curves.

On the full skin, a commuter would need 24.1 m^2 instead of 14.4 m^2, and a downtown street parker 51.3 m^2, more than the roof, hood and sides of the vehicle can hold. Put the other way, the 11.6 m^2 skin covers 17.8 km/day for the commuter against 37.5 km/day in the baseline. Putting all 11.6 m^2 on near-horizontal surfaces would raise F_{geo}, but the roof holds

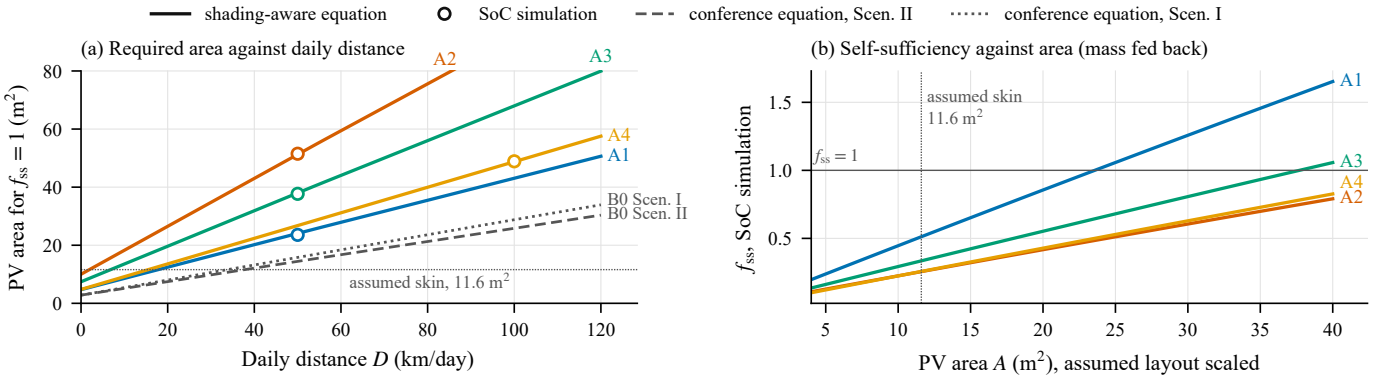


Fig. 7. (a) Area for $f_{ss} = 1$ against daily distance from (8) for each archetype (full skin, $\kappa_{pv} = 0.80$), with the hour-by-hour simulation (circles) and the conference equation (2) (gray). (b) f_{ss} against area in the simulation, with the vehicle mass fed back.

TABLE IV
LOSS FACTORS AND REQUIRED AREA (FULL SKIN, $\kappa_{pv} = 0.80$)

	F_{geo}	F_{sh}	F_{mm}	A_{req} (m^2)		Error	D_{max}
				(8)	simul.	(%)	(km/day)
A1	0.80	0.79	0.94	24.1	23.6	2.1	17.8
A2	0.80	0.40	0.87	51.3	51.5	−0.5	1.9
A3	0.80	0.58	0.82	38.0	37.7	0.8	6.8
A4	0.81	0.74	0.98	48.8	48.9	−0.3	15.5

Simul.: hour-by-hour simulation solved for $f_{ss} = 1$. Error = (Eq. (8) − simul.)/simul. B0 for comparison: $A_{req} = 14.4$ (Scen. II) and 28.8 m^2 (Scen. I). D_{max} at 11.6 m^2.

only 4.4 m^2; the sweep table in the repository gives f_{ss} for every area, allocation, κ_{pv} and archetype.

VIII. VALIDATION OF THE TWIN

Without measured on-vehicle irradiance, the twin is cross-verified, not physically validated; Table V lists every check with its target and status. Unity 6 with Cesium for Unity imported the same meshes around R-DENSE, replayed its 30 traces on the four dates (120 trip-dates) and cast PhysX rays from every sample once per second; Fig. 9 shows one moment. Its surface lies within 0.14 m of the lidar at 5 survey marks. Unity and the raster engine agree on 99.1% of 15.5 million sample-seconds. Trip lit fractions agree within 2 percentage points (pp) on 119 of 120 trip-dates, so this target was not met; the largest difference is 2.37 pp (Fig. 8a). Embree rays on the same meshes match Unity to 0.02 pp, so the remaining disagreement comes from how the surface is represented, not from the ray tracing: the raster treats each pixel as a flat-topped column, while the mesh rounds roof edges. The ±1° band around the horizon line holds 3.2% of sample-seconds but 65% of disagreements; outside it every trip agrees within 0.65 pp.

Mapillary panoramas [34] within 10 m of R-DENSE and R-MIXED and at least 250 m apart give 8 test points. Their sky segmentation gives SVF with the engine's cosine weighting. All 8 give a mean absolute error (MAE) of 0.171. One panorama was taken on an overpass deck that the route passes beneath, where the engine correctly sees the deck; without it,

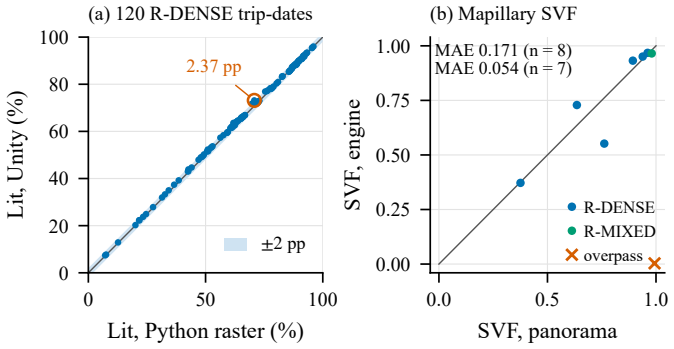


Fig. 8. (a) Trip lit fraction, Unity against the raster engine; the band is ±2 pp and the circled trip has the largest difference. (b) SVF from Mapillary panoramas against the engine at 2 m.

TABLE V
VERIFICATION AND VALIDATION METRICS

Check	Result	Target	Status
Engine tests	40/40	all	met
Raster vs. mesh, beam state	99.6%	≥ 98%	met
Raster vs. mesh, SVF mean $\lvert\Delta\rvert$	0.008	< 0.02	met
SOLWEIG, shadow pixels	98.0–99.3%	≥ 95%	met
SOLWEIG, parking sites	≥96.7%	≥ 95%	met
Unity vs. raster, sample-seconds	99.1%	≥ 97%	met
Unity vs. raster, trips within 2 pp	119/120	all	not met
Mapillary SVF MAE, $n = 8$ / 7	0.171 / 0.054	reported	–
Energy closure, all steps	1×10^{-14}	$\leq 10^{-3}$	met
Design equation vs. simulation	2.6%	≤ 3%	met

Engine tests: sun position, transposition, shadow length, canyon SVF. Beam: 1000 points, 43 suns. SOLWEIG: 12 times, 36 site maps. Mapillary $n = 7$ leaves out the overpass panorama. The missed trip target is one trip at 2.37 pp inside the ±1° horizon band.

the remaining 7 give MAE 0.054 and $r = 0.92$ (Fig. 8b). With so few points this is a plausibility check.

IX. DISCUSSION AND LIMITATIONS

Where the car parks matters more than where it drives: the commuter keeps 0.512 because it parks in the open by day, while the downtown street parker keeps 0.256, and for both, parking causes most of the shading loss (92% and 95%). For design, (8) turns a site survey into three multiplicative factors that can be estimated from shade maps without a full



Fig. 9. The Unity twin on R-DENSE northbound at 08:03:52 MST on 21 December (08:00 departure, medium traffic). The analysis geometry casts the shadows in shadows-only mode under Google Photorealistic 3D Tiles. Dots on the skin are the 0.25 m samples: yellow lit, blue in building shadow, green in tree shadow. The inset at right gives the lit share and power of each facet. Imagery: Google; Cesium.

simulation, and the error of the closed form is smaller than the spread between archetypes.

The method transfers to other cities: London's weather is already in Table I, and the geometry needs national lidar and the same building and street sources. A lower sun and narrower streets would likely raise the parking and orientation losses there, so the London baseline of 0.517 is an upper bound.

The limitations are as follows. The skin layout is an assumed design; side panels dominate F_{geo}, so a roof-heavy design would lose less to orientation. Mismatch depends on the module layout (three diodes per facet module and 0.25 m cells); more diodes or per-substring tracking would reduce it [17]. Trees use one transmissivity and the lidar canopy of 2020–2021, without seasonal leaf change. Traffic is simulated and the arterials stay below saturation, so queues in shade are underrepresented. The weather is a typical year; the shading and SVF were checked against models and imagery, and no irradiance was measured on a vehicle [6]; and 114 smaller building changes along the routes were not reviewed.

X. Conclusion

A digital twin of central Phoenix shows that the usual horizontal, unshaded roof overstates the solar self-sufficiency of a heavy VIPV vehicle: the four archetypes lose 38% to 69% of the baseline value. Starting from 0.832 at 50 km/day, facet orientation and shade leave 0.512 for a suburban commuter and 0.256 for a car parked at a downtown curb, and grid charges rise from 5 to 13–20 a year. Parking shade is the largest loss for the two downtown parkers and second to orientation for the commuter and the fleet vehicle; driving through street canyons costs at most 0.017. A shading-aware design equation reproduces the full simulation within 2.6% and gives the required area for a new site from three factors.

Code and Data Availability

The pipeline, configuration, frozen route manifest and result tables are at https://github.com/jarvisxys/solaris-vipv, and the release used for this paper (v2.0.0-journal) is archived on Zenodo, doi: 10.5281/zenodo.23896001 [35]; the conference toolkit is at https://github.com/Anandsagar18/Solaris [2]. Lidar, NAIP and NSRDB data are public and fetched by the pipeline; Overture and OpenStreetMap data are under ODbL and Mapillary imagery under CC BY-SA 4.0.

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